

Email: l.demonte@ed.ac.uk

Google Scholar: [Publications](#)

GitHub: [Repositories](#)

My PhD developed geometric approaches to multivariate extremes. Going forward, my research focuses on generative methods for spatial extreme events, in the context of environmental hazards and compound climate extremes.

CURRENT POSITION

King Abdullah University of Science and Technology (KAUST)

Aug. 2026 – Present

Incoming Postdoctoral Researcher, currently Remote Consultant.

EDUCATION

PhD Statistics, University of Edinburgh, United Kingdom

Sep. 2022 – Jun. 2026

Supervisors: Dr. Ioannis Papastathopoulos, Prof. Gabriele C. Hegerl

Thesis: *Geometric considerations for generative modelling of multivariate extremes*

Outcome: Successfully defended on June 29, 2026, with no corrections required.

Visiting PhD student, KAUST, Saudi Arabia

Feb. 2024 – May 2024 and Nov. 2025

Supervisor: Prof. Raphaël Huser

MSc Mathematics and Statistics, McGill University, Canada

Sep. 2020 – June 2022

Supervisor: Prof. Christian Genest

Thesis: *A spatial Gamma-Gamma model for the statistical postprocessing of ensemble weather forecasts*

BSc Mathematics and Computer Science, McGill University, Canada

Sep. 2017 – May 2020

PUBLICATIONS

Research

- De Monte, L. & Papastathopoulos, I. (2026), *A directional Poisson point process for univariate extremes*, [Statistics & Probability Letters](#) **235**, 110722.

- De Monte, L., Huser, R., Papastathopoulos, I., and Richards, J. (2025) *Generative modelling of multivariate geometric extremes using normalising flows*, [arXiv](#) preprint.

- Papastathopoulos, I., De Monte, L., Campbell, R. & Rue, H. (2023) *Statistical inference for radially-stable generalised pareto distributions and return level-sets in geometric extremes*, [arXiv](#) preprint.

Book chapter

- Auld, G., De Monte, L. & Papastathopoulos, I. (2026) Time series in extremes. In de Carvalho, M., Huser, R., Naveau, P., and Reich, B. J., editors, *Handbook on Statistics of Extremes*. Chapman & Hall/CRC, Boca Raton, FL.

Other

- Richards, J., De Monte, L. Review of “Risk Revealed: Cautionary Tales, Understanding and Communication” by Paul Embrechts, Marius Hofert, and Valérie Chavez-Demoulin. [JABES](#) (2024).

- Genest, C. and De Monte, L. (2021) *À la recherche de la Rondurie*, [Accromath](#), Volume 16 p.24-29.

Theses

- De Monte, L. *Geometric considerations for generative modelling of multivariate extremes*. PhD thesis. [The University of Edinburgh](#), 2026.

- De Monte, L. *A Spatial Gamma-Gamma Model for the Statistical Postprocessing of Ensemble Weather Forecasts*. Master’s thesis. [McGill University](#), 2022.

PROFESSIONAL EXPERIENCE

King Abdullah University of Science and Technology (KAUST)

Aug. 2026 – Present

Remote consultant

- Developing machine learning methods for the spatial modelling of environmental extreme events.

Hydro-Québec Research Institute (IREQ)

Jan. 2022 – Aug. 2022

Intern and Consultant

- Research partnership to develop methods for the statistical post-processing of weather forecasts, specifically of extreme meteorological events.

- Development and automation of the official DataHub credit card database.
- Statistical modeling of credit card use and behaviours.

AWARDS AND FUNDING

School of Mathematics, University of Edinburgh

Sep. 2023

- Full PhD Studentship: £27,600 tuition and £20,800 stipend per year

Mitacs Canada

Jan. 2022

- Funding for research on extreme value modeling in partnership with Hydro-Québec (CAD \$30,000)

TEACHING AND MENTORING

Teaching Assistantships (University of Edinburgh)

Tutoring workshops. Marking of assignments and examinations. List of courses:

- MATH10093 – Statistical Computing Jan. – May 2023, Jan. – May 2025, and Jan. – May 2026
- MATH08065 – Computing and Numerics Jan. – May 2026
- MATH10096 – Applied Statistics Jan. – May 2025 and Jan. – May 2026
- MATH11199 – Python Programming Sep. – Dec. 2025
- MATH08066 – Probability Sep. – Dec. 2023 and Sep. – Dec. 2025
- MATH10064 – Multivariate Data Analysis Jan. – May 2025
- MATH11176 – Extended Statistical Programming Sep. – Dec. 2023
- MATH11187 – Generalised Regression Models Sep. – Dec. 2023
- MATH08051 – Statistics Jan. – May 2023

StatsBase Statistics Helpdesk (University of Edinburgh)

Nov. 2025

Providing help to MSc students in Statistics for the Statistical Methodology and Theory of Statistical Inference courses.

Teaching Assistantships (McGill University)

Preparation of tutorial sessions, office hours for course-related questions. Marking of midterm and final exams, supervision of assignment marking. List of courses:

- MATH 324 – Statistics Jan. 2022 – May 2022
- MATH 323 – Probability Sep. 2021 – Dec. 2021
- MATH 203 – Principles of Statistics Jan. 2021 – May 2021

Statistics Helpdesk (McGill University)

Sep. 2020 – Dec. 2021

Providing help to undergraduate students in Statistics and Probability. Theoretical and programming guidance for courses in the range MATH 2xx–5xx.

Undergraduate Project Supervision (McGill University)

Guidance of three students towards the completion of an undergraduate project in extreme value theory.

- MATH 470 – Honours Research Project Jan. 2022 – May 2022
- MATH 410 – Majors Project Jan. 2021 – May 2021 and Sep. 2021 – Dec. 2021

ACADEMIC SERVICE AND LEADERSHIP

Postgraduate representative, School of Mathematics (University of Edinburgh)

Representative of the postgraduate students of the *School of Mathematics*

Sept. 2024 – Oct. 2025

- Intermediary between the students and the School administration.
- Organisation of social events within the School.
- Representing the *School of Mathematics* at the *Martingale PhD Navigator Programme* events for future scholars (December 2024 and 2025).

Scientific article reviewing

Journal of Statistics and Computing.

PRESENTATIONS AND POSTER SESSIONS

Invited

- Generative AI Modelling for Extreme Events (GAME), Bologna, Italy: *Multivariate extremal index and clustering in multivariate extremes: A case study of Dutch hydrological extremes.* Jun. 2026
- CfS Annual Conference 2026, Edinburgh, United Kingdom: *Multivariate extremal index and clustering in multivariate extremes: A case study of Dutch hydrological extremes.* Jun. 2026
- Generative AI Modelling for Extreme Events, University of Edinburgh Generative AI Laboratory, Edinburgh, United Kingdom: *Generative modelling of multivariate geometric extremes using normalising flows.* Jun. 2025
- Statistics seminar, University of Exeter, Exeter, United Kingdom: *Geometric extreme value theory – A normalising flows approach.* Feb. 2025
- CMStatistics, London, United Kingdom: *Geometric extreme value theory – A normalising flows approach.* Dec. 2024
- CMStatistics, Berlin, Germany: *Bayesian inference for radially-stable distributions.* Dec. 2023
- BIRS-IMAG Modern Statistical and Machine Learning Approaches for High-Dimensional Compound Spatial Extremes, Granada, Spain (short presentation): *Geometric inference for Hüsler–Reiss random vectors.* May 2023
- Séminaires de sciences de la décision HEC Montréal, Montréal, Canada: *Multivariate extremes – A geometric Bayesian inference approach.* Sep. 2023

Contributed

- 14th International Conference on Extreme Value Analysis, University of North Carolina Chapel Hill, United States: *Generative modelling of multivariate geometric extremes using normalising flows* Jun. 2025
- 15th International Meeting on Statistical Climatology, Toulouse, France: *Flood risk modelling using geometric extreme value theory.* Jun. 2024
- CfS Annual Conference 2024, Edinburgh, United Kingdom (poster): *Multivariate radial Pareto distributions: a geometric approach to the statistical modelling of multivariate extremes.* Jun. 2024
- 13th International Conference on Extreme Value Analysis, Milan, Italy (poster): *Bayesian approach to geometric inference for multivariate extremes.* Jun. 2023

SOFTWARE (OPEN-SOURCE)

- Python code [geometricExtremesNF](#): Creator and maintainer. Based on De Monte et al. (2025)
- R package [geometricExtremes](#): Creator and maintainer. Based on Papastathopoulos et al. (2023).
- R code [TSExtremes](#): Collaborator. Based on Auld et al. (2026).

TECHNICAL SKILLS

Languages: R, Python, Julia, Bash, SQL, LaTeX.

High-performance computing: Eddie (University of Edinburgh), Ibex (King Abdullah University of Science and Technology), CASIR (Hydro-Québec).

Cloud computing: Microsoft Azure (National Bank of Canada).

LANGUAGES

Bilingual: English, French.